Indigestion and ‘heartburn’: Nausea and vomiting Gastric motility problems Benign esophageal lesions

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Outline of lecture

- Approach to indigestion and heartburn
- Approach to nausea and vomiting

- Overview of gastric motility problems
- Overview of benign esophageal lesions

Approach to indigestion and heartburn

Heartburn

- Typically refers to ***gastroesophageal reflux disease (GERD)***
- A ***burning feeling*** rising from the stomach or lower chest up towards the neck
- May include varying symptoms:
  - Acid regurgitation
  - ***Sour stomach***
  - Bitter belching
  - Waterbrash
- ***Regurgitation:*** effortless return of gastric contents upward toward the mouth, often accompanied by an acid or bitter taste
- Most frequently noted within 1 hour after eating or activities that ***raises intra-abdominal pressure or lying down***

Indigestion/ dyspepsia

- Pain or discomfort centered in the upper abdomen
- May include varying symptoms:
  - Epigastric pain/ burning
  - ***Postprandial fullness***
  - ***Early satiety***
  - Epigastric bloating
  - (Anorexia/ nausea)
  - (Sometimes patients refer to heartburn or regurgitation)

Overlap between symptoms of gastroduodenal origin and symptoms of presumed esophageal origin – area of controversy

Definition of Gastro-Esophageal Reflux Disease (GERD)

The condition in which the reflux of gastric contents into the esophagus ***results in symptoms and/or complications***

Reflux of gastric acid (can be other contents e.g. bile juice, gas),
i.e., GER, itself is not a disease (physiological)

Pathophysiology of GERD

- GERD is a spectrum of disease resulting from *imbalance* between *anti-reflux barriers vs. aggravating factors*

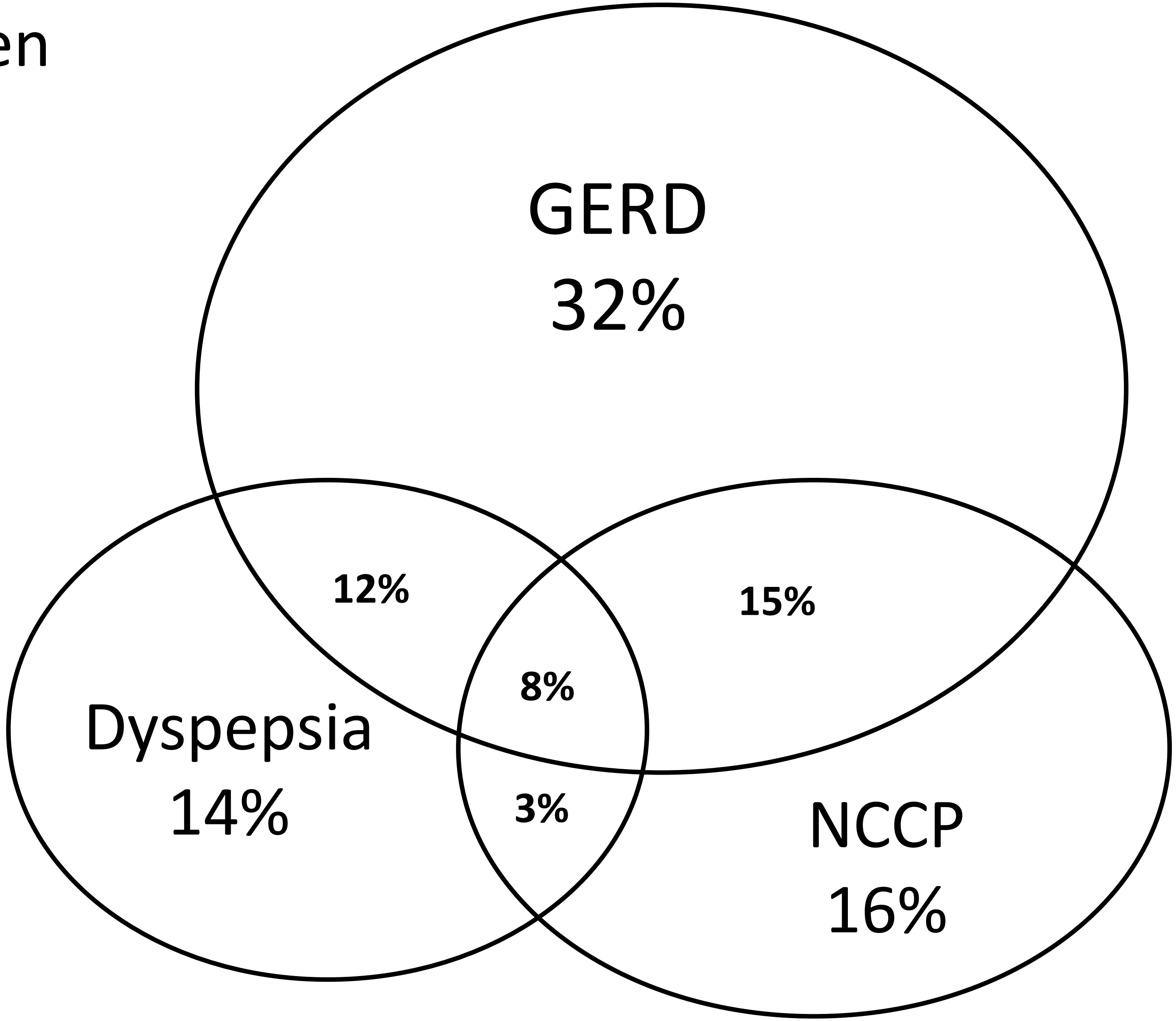
- Anti-reflux barriers:
  - LES (resting pressure ~10mmHg) intra-abdominal location of LES
  - the hiatus (diaphragmatic crus + phrenoesophageal ligaments)
  - Oblique entrance of esophagus into the stomach → sharp angle on the greater curvature (the angle of His) – flap valve effect

- Aggravating factors:
  - tLESR (transient LES relaxation) ← stimulated by proximal gastric distension
  - Hiatus hernia
  - Impaired esophageal emptying/ peristalsis
  - Hypotensive LES (post-POEM)
  - Acid pocket (unbuffered by meal, remain low pH)
  - Excessive gastric acid (e.g., hypergastrinaemia ← Hp+ve antral gastritis, post-Hp eradication for corpus-predominant gastritis, Zollinger Ellison syndrome; delayed gastric emptying)

GERD – Clinical manifestations

- ***Typical GERD symptoms***
  - Post-prandially, after large meal
  - Some food types trigger more symptoms: spicy food, citrus product, fat, chocolates, alcohol
  - Supine, right decubitus posture
- Atypical chest pain
- Extra-esophageal symptoms
  - Asthma, chronic cough, sleep disturbance, laryngo-pharyngeal reflux, hoarseness of voice, throat tightness, dental erosion
- ***Endoscopic classification***: erosive vs. non-erosive
- ***Complications***
  - Esophagitis, esophageal ulcer and stricture
  - Barrett's esophagus
  - Adenocarcinoma of esophagus

Overlap between GERD, NCCP & dyspepsia



NCCP: non-cardiac chest pain

Wong WM, AP&T 2003;18:595

Differential diagnosis of GERD

- Achalasia
- Zenker’s diverticulum
- Gastroparesis
- Angina pectoris
- Causes of dyspepsia
- Other causes of esophagitis: pill-induced, infection, radiation injury

Associated conditions of GERD

- Pregnancy (incidence 30-80%)
- NG tube
- Scleroderma (38% have erosive disease)
- Multiple endocrine neoplasia (MEN)
- Obesity

Diagnosis of GERD

cost-effective

- Proton pump inhibitor test (***PPI test***)
  - Accurate and cost-effective in patients with symptoms suggestive of GERD and NCCP.
  - PPI empirical trial can be considered as initial diagnostic step in patients with the disease spectrum of GERD.
  - Try standard PPI for 8 weeks.
- ***Diagnostic questionnaire***
  - Validated Chinese questionnaire, with 7 questions

Subjective

- ***24 hour esophageal pH*** monitoring
- ***Upper endoscopy***

Objective

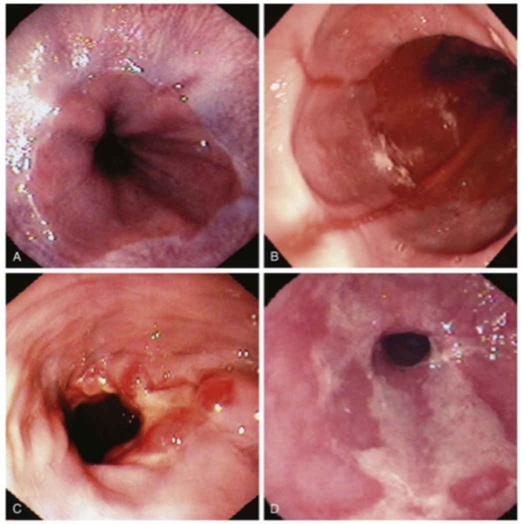
CHINESE (Cantonese) GERDQ

SYMPTOM ASSESSED	
• Frequency of “heartburn”	*火燒心*
• Severity of “heartburn”	
• Frequency of feeling of acidity in stomach	*胃酸過多*
• Severity of feeling of acidity in stomach	
• Frequency of acid regurgitation	*反流/胃酸倒流*
• Severity of acid regurgitation	
• Frequency of “use of antacids”	

Wong WM et al. AP&T 2003; 17: 1407

Role of endoscopy in GERD

1. ***Diagnosing erosive GERD*** (~30% of all GERD; i.e. 70% GERD is NERD)
   - However, only 20-60% patients with abnormal esophageal pH have esophagitis on endoscopy (the rest non-erosive reflux disease – NERD). ***Normal endoscopy finding does not exclude GERD***

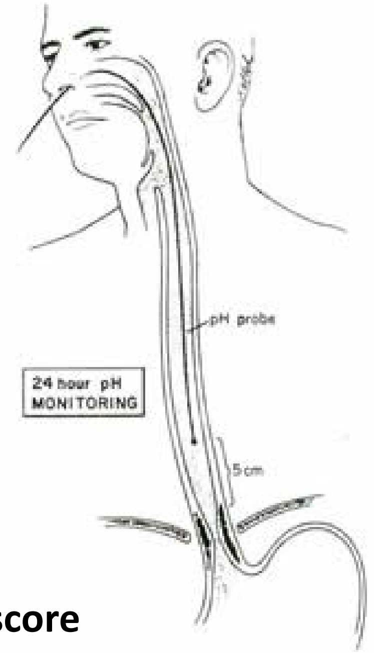


Los Angeles Endoscopic Classification system for esophagitis	
A	≥ 1 mucosal breaks confined to folds, ≤5mm long
B	≥1 mucosal breaks confined to folds, >5mm long
C	≥1 mucosal breaks continuous between tops of ≥2 mucosal folds, <75% circumference
D	≥1 mucosal breaks continuous between tops of ≥2 mucosal folds, ≥75% circumference

Note: the procedure is invasive, expensive and uncomfortable

2. ***Look for cause*** (hiatus hernia, EoE, antral gastritis, associated PUD)
3. ***Look for complications*** (esophageal strictures, Barrett's esophagus, adenocarcinoma)
   - Alarm symptoms (e.g. weight loss, dysphagia, odynophagia, GI bleeding)

Portable pH recorder and pH sensors



Parameters in the DeMeester score

Time in which pH <4.0

Parameters	Asymptomatic controls ($n = 15$)	Normal value
Recumbent period	0.286% ± 0.467	<1.2%
Total period	1.478% ± 1.381	<4.2%
No. episodes > 5 min	0.6 ± 1.241	3 or less
Longest episode	3.866 min ± 2.689	<9.2 min
Upright period	2.33% ± 1.975	<6.3%
No. total episodes	20.6 ± 14.773	<50

Johnson LF and DeMeester TR. J Clin Gastroenterol 1986

- Catheter with 2 antimony pH electrodes.
- A study is abnormal if the total percentage of time with pH less than 4.0 is ≥4.2% in the distal esophagus (DeMeester score).
- Often referred as the 'gold standard' in diagnosing GERD.
- The sensitivity of 24-hr esophageal pH monitoring in diagnosing GERD with esophagitis is about 90%.
- However, the sensitivity for NERD is much lower.
- The procedure is unpleasant to the patient.
- May not be freely available to community physician.

New esophageal pH monitoring technique

The Catheter-Free pH System (BRAVO™)

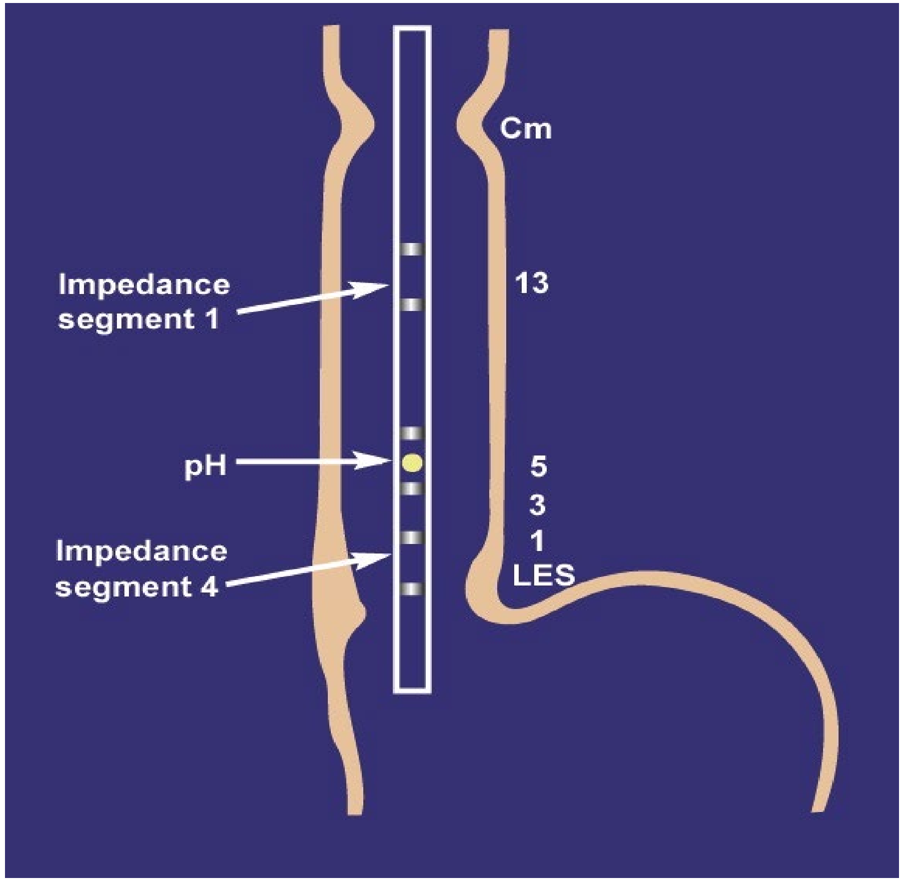
The pH capsule

The capsule transmits pH data using radio frequency signals to a small pager size receiver that is worn by the patient

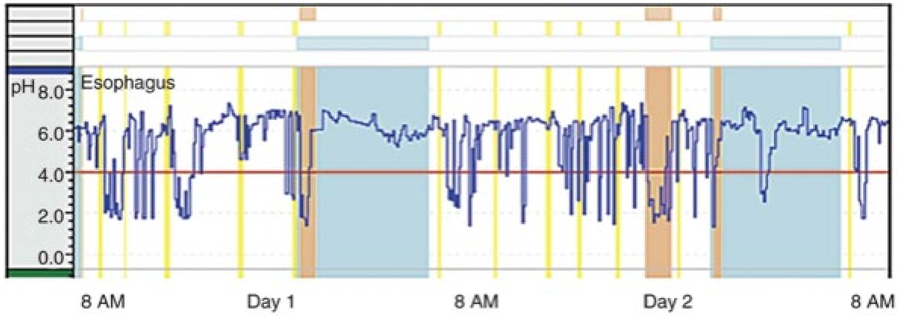
Combined Multichannel Impedance and pH Catheter

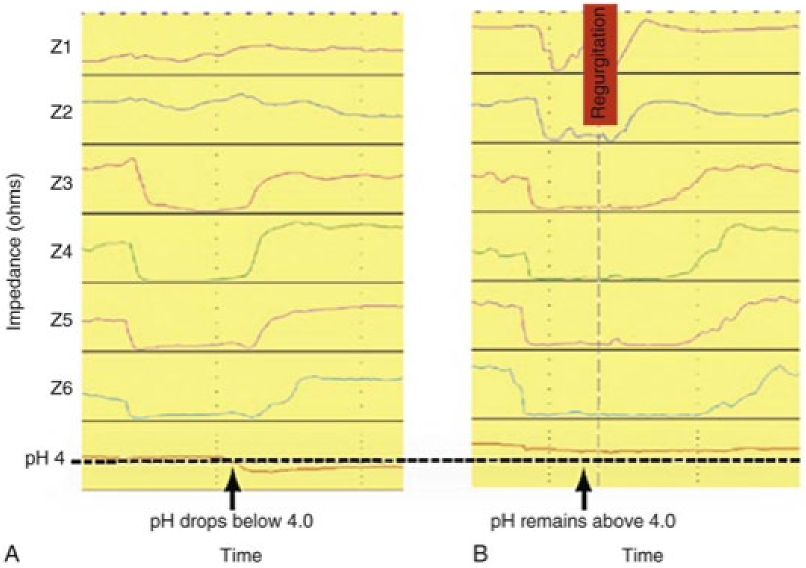
Contents of esophageal reflux:

- Acidic – most common
- Non-acidic: bile, bolus of food, swallowed liquid, air



Sifrim *et al.*, Gastroenterology 2001





Tracings from a 48-hour esophageal pH (top panel) + multichannel impedance-pH studies (lower panel).
Yellow lines = meal/ drinks
Blue lines = supine
Orange lines = symptoms + pH drop

GERD: Aims of treatment

- Confirm diagnosis of reflux disease
- Relief of reflux symptoms
- Reassurance of patients
- Healing of esophagitis, if present
- Improve quality of life
- Prevent long term complications

Therapeutic options in GERD

- ***<u>Lifestyle modification</u>***

Table 3. **Recommendations based on results of a review of studies involving lifestyle modifications**

Lifestyle modification	Strength of scientific evidence	Pathophysiologically conclusive?	Recommendable?
Avoid fatty meals	Equivocal	Equivocal	Yes
Avoid carbonated beverages	Moderate	Yes	Yes
Select decaffeinated beverages	Equivocal	Equivocal	Not generally
Avoid citrus	Weak	Yes	Yes, if citrus triggers symptoms
Eat smaller meals	Weak	Yes	Yes
Lose weight	Equivocal	Equivocal	Yes[a]
Avoid alcoholic beverages	Weak	Mechanisms not understood; different alcoholic beverages have different effects	Not generally
Stop smoking	Weak	Yes	Yes[a]
Avoid excessive exercise	Weak	Yes	Yes
Sleep with head elevated	Equivocal	Equivocal	Yes
Sleep on the left side	Unequivocal	Yes	Yes

[a]Obesity and smoking seem to be risk factors for cancer of the distal esophagus.

Therapeutic options in GERD

- Medications
  - Antacids
    - On demand symptom relief with little evidence
  - H2 receptor antagonist
    - Main role as add-on therapy for bedtime breakthrough GERD symptoms despite PPI
    - Tachyphylaxis
  - ***<u>Proton pump inhibitors</u>***
    - Superior relief of heartburn/ regurgitation + healing of esophagitis c.f. H2RA
    - Best to be taken 30-60 mins before meal (PPI can only bind to proton pumps that are actively secreting acid, and meals stimulate proton secretion)
    - Try to discontinue long-term therapy as much as possible
    - Maintenance therapy
      - Long term: for erosive esophagitis (LA grade C/D), peptic stricture, Barrett's esophagus
      - On-demand/ intermittent: NERD patients with severe GERD symptoms responsive to PPI

No role: Baclofen, prokinetics, sucralfate (except during pregnancy)

Safety of long-term PPI

- Known effects: fundic gland polyps (mostly benign), acute interstitial nephritis
- Many association studies reported numerous potential adverse conditions (e.g. intestinal infections, pneumonia, gastric cancer, osteoporosis-related bone fractures, CKD, stroke, dementia, hepatic encephalopathy in cirrhotic patients, hypomagnesemia)
- But ***NO cause-and-effect studies confirming causation***
  - ***except enteric infections other than Clostridium difficile***; OR 1.33 (95% CI 1.01-1.75)
- Bottom-line:
  - PPIs are the most effective medical treatment for GERD
  - Although cannot exclude the possibility that PPIs do increase risk of the reported adverse events, the benefits > theoretical risks for patients whom are deemed indicated for long-term PPI
- Guidelines do not recommend routine additional calcium or vit D supplement/ DEXA/ monitoring serum B12/ Cr

Moayyedi P et al. Gastroenterology 2019; 157: 682-691)
Mak LY et al. Hepatol Int 2020; 14:299-301

Table 5. Major putative adverse effects of chronic PPI therapy

Putative adverse effect	Meta-analysis reference numbers	HR[a] or OR[b] (95% CI) found in recent RCT (94)	Major proposed mechanisms
Cardiovascular events (All) MI Stroke Cardiovascular death	(237–240)	1.04[a] (0.93–1.15) 0.94[a] (0.79–1.12) 1.16[a] (0.94–1.44) 1.03[a] (0.89–1.20)	PPIs block metabolism of ADMA, which accumulates and inhibits NO synthase, thus blocking endothelial production of NO needed for vascular homeostasis
Cardiovascular events in patients on clopidogrel[c]	(241–260)	NA	PPIs are metabolized by the same enzyme needed to activate clopidogrel (CYP2C19), so concomitant use of these drugs might decrease the antiplatelet effect of clopidogrel
Kidney disease (All) AIN Chronic kidney disease	(261–265)	NA NA 1.17[b] (0.94–1.45)	AIN develops as an idiosyncratic drug reaction and progresses to chronic kidney disease
Enteric infection (other than *Clostridium. difficile*)	(266,267)	1.33[b] (1.01–1.75)	Reduced gastric acid enables ingested enteric pathogens to survive passage through the stomach
C. difficile	(268–276)	2.26[b] (0.70–7.34)	Reduced gastric acid enables survival of ingested *C. difficile* vegetative forms and prevents conversion of salivary nitrite to ROS that suppress *C. difficile* spores; PPIs may enhance *C. difficile* toxin expression and cause microbiome alterations that promote *C. difficile* colitis
SIBO	(277,278)	NA	Reduced gastric acid enables increased bacterial colonization of the UGI tract
Spontaneous bacterial peritonitis in patients with cirrhosis	(279–283)	NA	Increased bacterial colonization of the UGI tract and PPI-induced increases in UGI tract permeability predispose to bacterial translocation; PPIs also might interfere with inflammatory cell functions that ordinarily would prevent infection
Pneumonia	(284–289)	1.02[b] (0.87–1.19)	Reduced gastric acid enables UGI tract colonization with pulmonary pathogens that can be aspirated; PPIs also might interfere with inflammatory cell functions that ordinarily would prevent infection
Dementia	(290–293)	1.20[b] (0.81–1.78)	PPIs block vacuolar H^+-ATPase needed to acidify microglial lysosomes, thereby preventing their degradation of cerebral amyloid-β peptide; PPI-induced B12 deficiency also might contribute to dementia
Bone fracture	(294–302)	0.96[b] (0.79–1.17)	Reduced gastric acid causes calcium malabsorption leading to decreased bone mineral density; PPIs might reduce bone resorption by blocking vacuolar H^+-ATPase in osteoclasts; PPIs cause hypergastrinemia that might cause parathyroid hyperplasia
Gastric atrophy	(303–305)	0.73[b] (0.40–1.32)	PPIs promote corpus-predominant *H. pylori* gastritis that results in gastric atrophy with loss of parietal cells
Gastric cancer	(306–308)	NA	PPIs promote gastric atrophy and inflammation in *H. pylori*–infected patients, resulting in intestinal metaplasia predisposed to malignancy; reduced gastric acid enables overgrowth of bacteria that convert dietary nitrates to potentially carcinogenic N-nitroso compounds; PPI-induced hypergastrinemia causes gastric epithelial cell proliferation that promotes carcinogenesis
Vitamin B12 deficiency	(309)	NA	Reduced gastric acid results in malabsorption of protein-bound cobalamin; gastric atrophy results in decreased intrinsic factor production
Hypomagnesemia	(310–312)	NA	PPI effects in elevating intestinal pH may interfere with magnesium absorption, perhaps because the affinity of the enterocyte magnesium transporter TRPM6/7 for magnesium decreases in a higher pH environment
All-cause mortality	(313)	1.03[a] (0.92–1.15)	Potentially all of above

What to do for patients with refractory GERD?

Table 1 Diagnostic possibilities for refractory reflux symptoms

Non-GORD	Delayed gastric emptying (common) Motility disorder: achalasia (common) Functional (common): normal reflux burden, no symptom/reflux correlation Aerophagia (less common) Rumination (less common) Eosinophilic oesophagitis (if dysphagia is present)
Weakly acidic/ non-acidic reflux	May occur after acid suppression in context of regurgitation due to mechanical failure (large hiatal hernia)
Insufficient acidic suppression	Dosing (common) Compliance (common) Zollinger-Ellison syndrome (rare) PPI resistance (less common)
Reflux sensitivity	Reflux burden is normal but patient has a clear symptom/reflux correlation. This is dependent on visceral hypersensitivity and hypervigilance

PPI, proton pump inhibitor.

Fock KM, *et al. Gut* 2016;**65**:1402–1415. doi:10.1136/gutjnl-2016-311715

Possible actions:

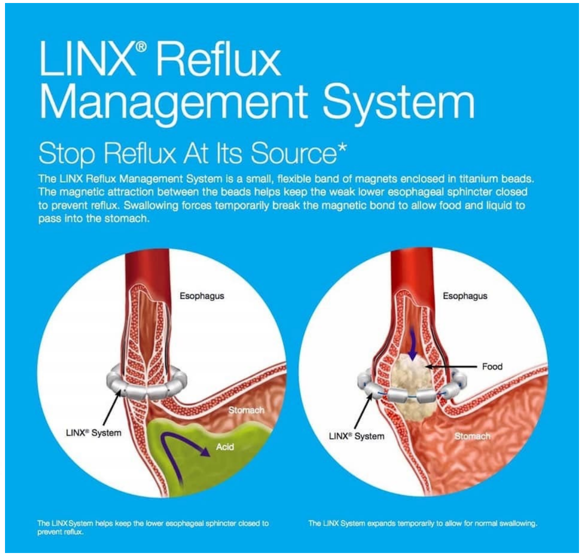
- Optimize PPI dose, timing, compliance?
- Try switch class of PPI (metabolism)
- Add nocturnal H2RA for night-time symptoms
- Add Alginate (e.g. Gaviscon) for post-dinner symptoms
- 24 hour pH monitoring +/- impedance study
- Anti-reflux surgery (confirmed with objective testing)
- Esophageal manometry
- Repeat OGD after OFF PPI for 2-4 weeks (drug holiday) to look for EE & obtain esophageal biopsies (EoE) – 1-8% refractory GERD

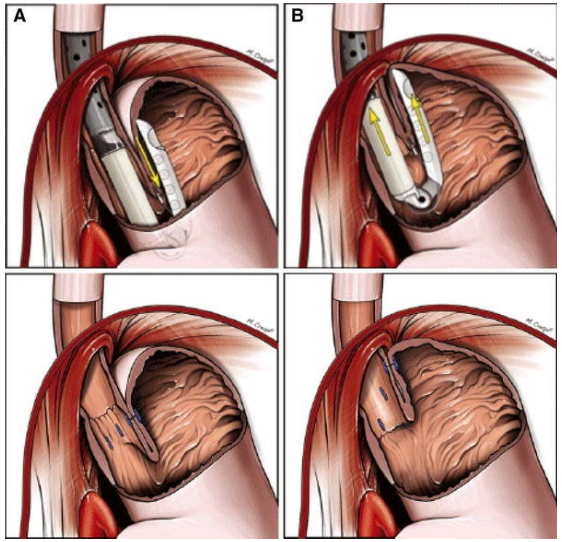
Therapeutic options in GERD

- ***<u>Surgical options</u>***
  - Anti-reflux surgery:
    - Laparoscopic fundoplication (Nissen)
    - Magnetic sphincter augmentation (MSA) - LINX
  - Trans-oral incisionless fundoplication (TIF)

Pre-op evaluation:
- Confirm the diagnosis of GERD
- Confirm the symptoms are (at least partially) PPI-responsive
- Exclude contraindications for anti-reflux surgery e.g. achalasia

Drawbacks:
- Surgical risks (~4%)
- Recurrence of GERD post-op (~20% long term)
- Post-operative dysphagia/ gas bloat/ inability to belch





Pearl J et al. Surgical Endoscopy 2017

Workup for uninvestigated indigestion/ dyspepsia

- Most prevalent cause: peptic ulcer disease, Hp infection, functional dyspepsia
- Significant organic cause to exclude: malignancy of upper GIT, hepatic-pancreatic-biliary cause

1. **<u>Upper endoscopy:</u>** when to offer?
   - Red flag symptoms (e.g. weight loss, anaemia, dysphagia, persistent vomiting)
   - Age ≥ 60

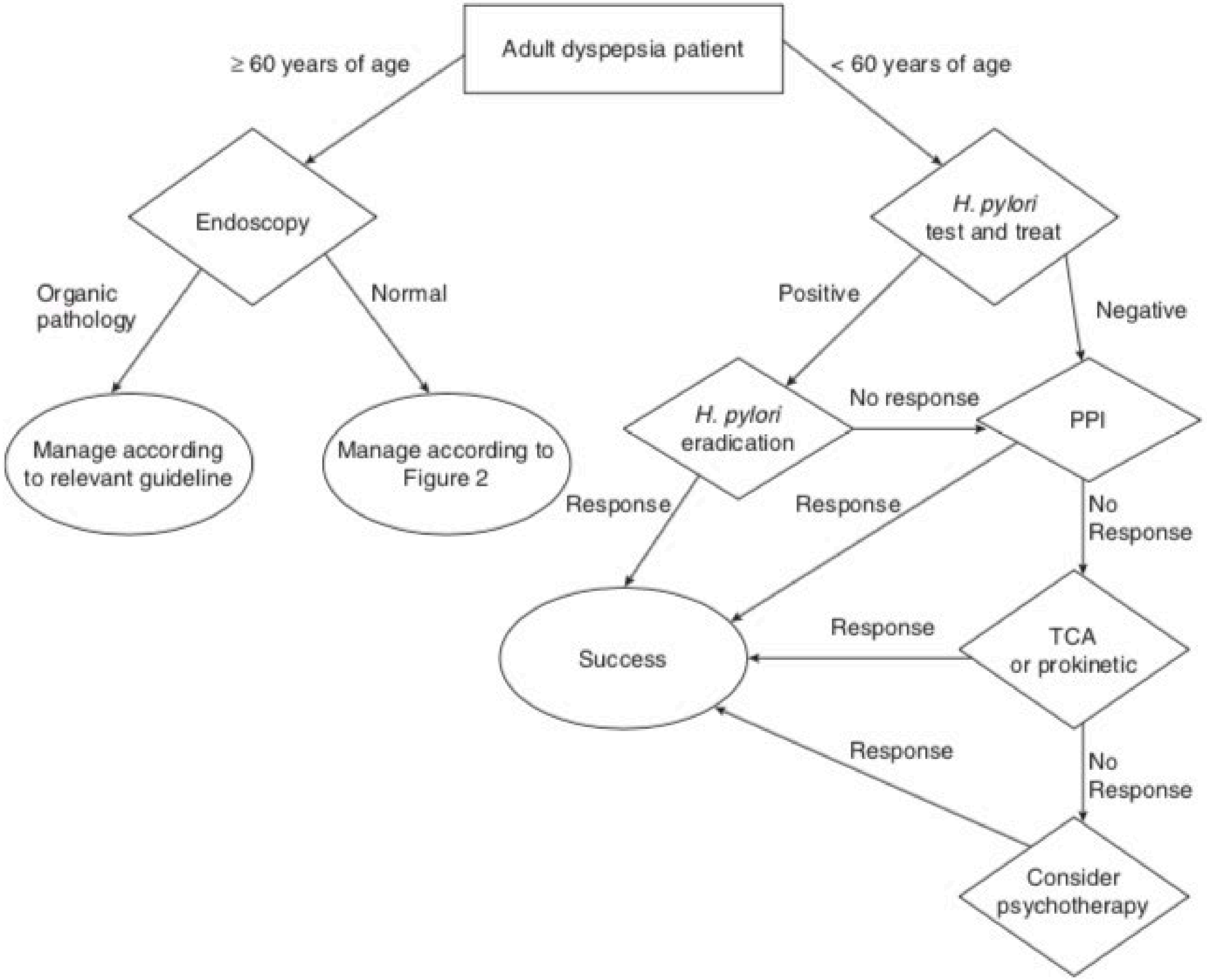
2. **<u>Test (for Hp) and treat approach</u>**
3. **<u>Empirical anti-secretory drug</u>**

If no response or recurrence of symptoms, may eventually need upper endoscopy

- Laboratory tests: CBC, LRFT, TFT: cost-effectiveness not established

2 & 3 maybe sequential; i.e. tested for Hp first, if –ve can give empirical PPI therapy or remain symptomatic after Hp eradication

Algorithm for the management of uninvestigated dyspepsia



Functional dyspepsia

- Dyspepsia with normal endoscopy
- Note that presence of Helicobacter pylori does not exclude FD
- And eradication of Hp can relieve symptoms in some patients with FD
- ***ROME IV criteria***

Rome IV Diagnostic Criteria for Functional Dyspepsia

Presence of ≥1 of:

✳️ Postprandial fullness (3 days per week)
✳️ Early satiety (3 days per week)
✳️ Epigastric pain (1 day per week)
✳️ Epigastric burning (1 day per week)

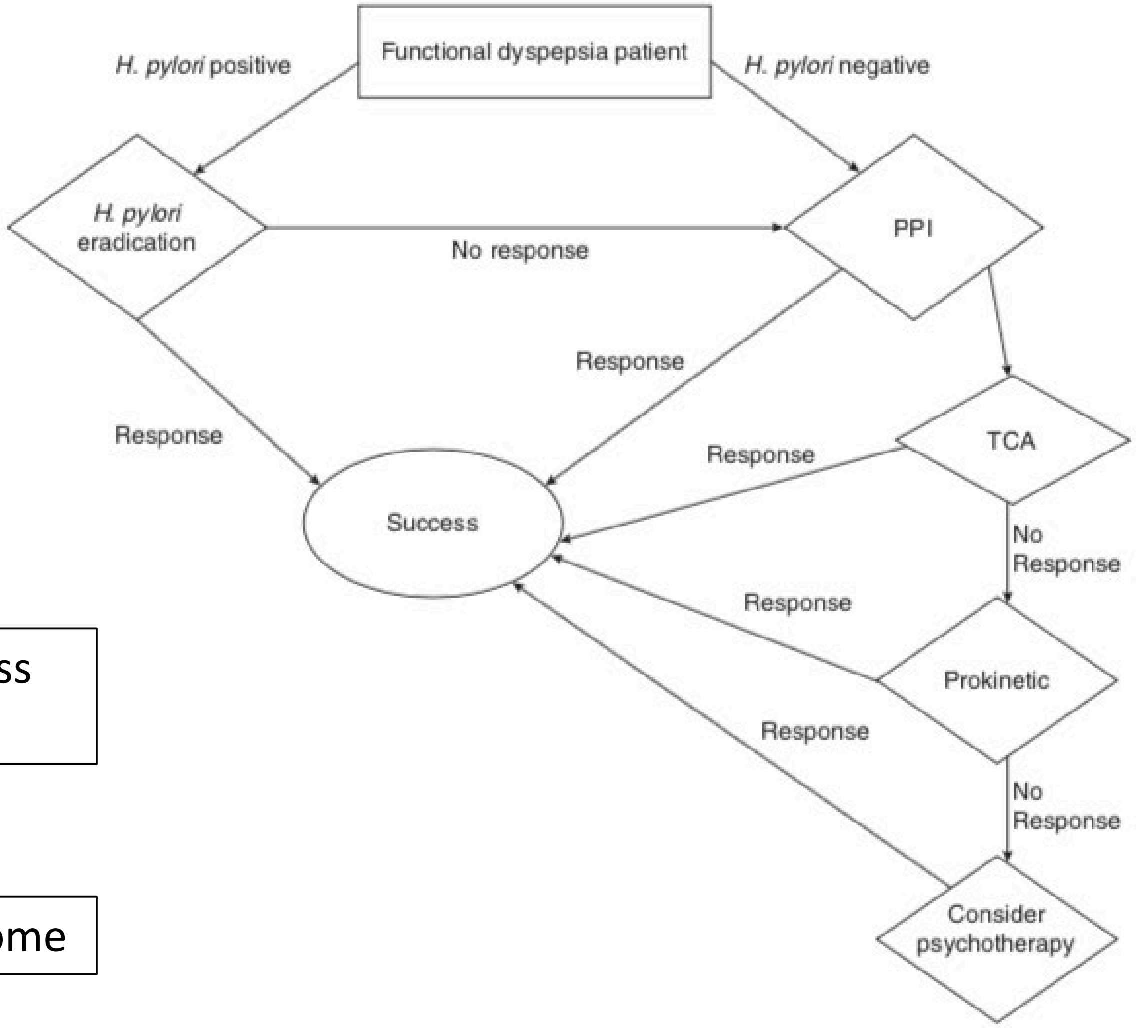
and

✅ No evidence of structural disease

Note: Criteria must be present for at least the past 3 months, with symptoms starting at least 6 months before diagnosis

Post-prandial distress syndrome

Epigastric pain syndrome



Stanghellini V et al. Gastroenterology 2016; 150:1380-92

Approach to nausea and vomiting

Nausea and Vomiting

- Nausea → retching → vomiting
  leading to the emetic reflex; but may occur separately

- **Nausea**: an unpleasant subjective sensation as a feeling of impending vomiting in the epigastrium or throat
- **Retching**: consists of spasmodic and abortive respiratory movement with the glottis closed
- **Vomiting**: partially voluntary act of forcefully expelling gastric or intestinal content through the mouth, and is a complex act that requires central neurologic coordination

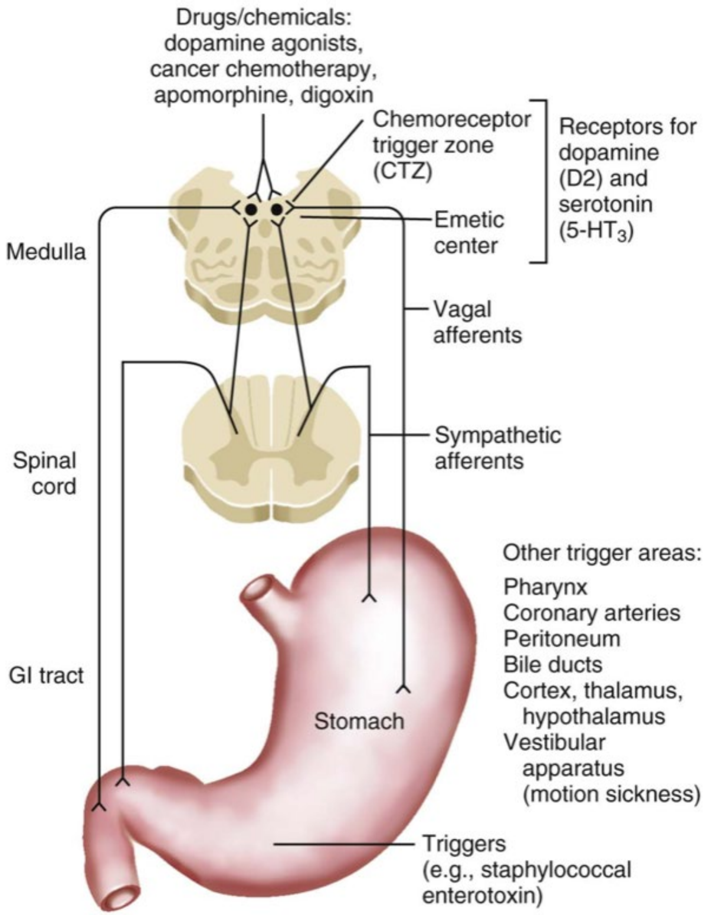
  vs ***Regurgitation***: effortless reflux of gastric contents into the esophagus but not associated with forceful ejection of contents

Pathophysiology of vomiting

- Emetic center is located in the medulla (dorsal portion of the lateral reticular formation): the solitary nucleus
- Afferent neural pathways carry activating signals from:
  - Pharynx
  - Stomach, small intestine, bile duct
  - Peritoneum
  - Heart
  - Testicles
  - CNS: cerebellum, vestibular system, brainstem, chemoreceptor trigger zone

GI tract (Pharynx; Stomach, small intestine, bile duct)

Non-digestive organs (Peritoneum; Heart; Testicles; CNS)



- Efferent pathway:
  1. cerebral cortex activation
  2. stomach: antral relaxation & inhibition of intestinal peristalsis
  3. Intercostal muscles and diaphragm: spasmodic contraction + closure of glottis
  4. Brisk muscular contraction, relaxation of LES, forceful retrograde peristalsis of jejunum +/- hypersalivation, cardiac arrhythmia

Causes of vomiting

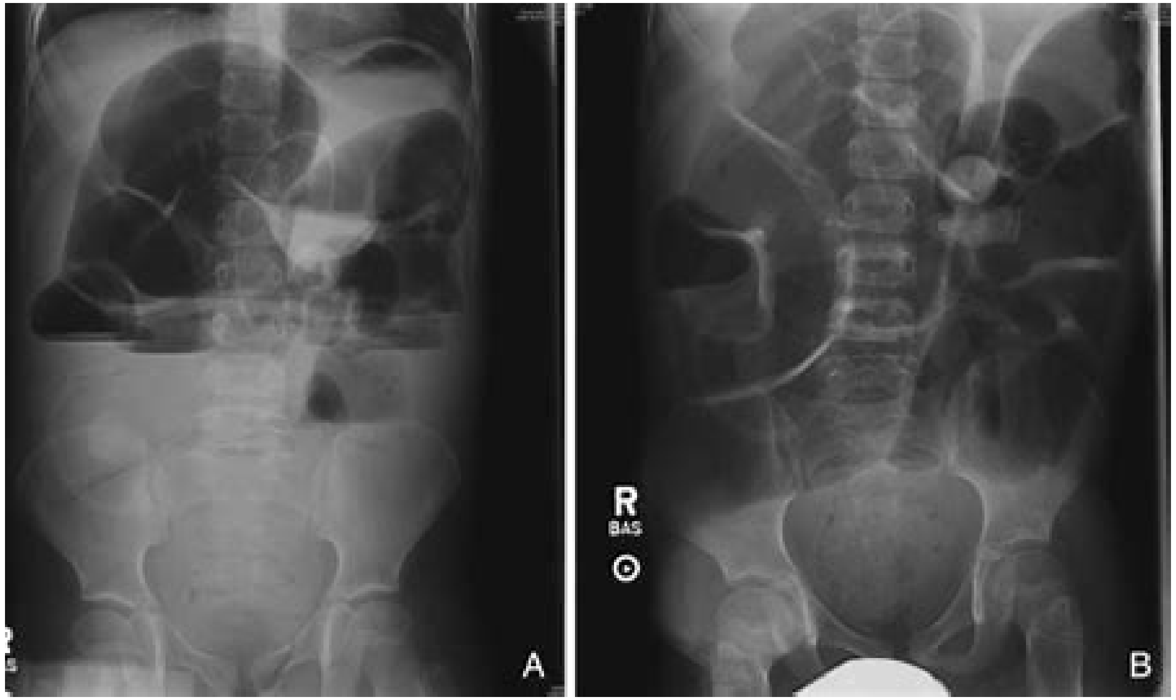
GIT

- Mechanical obstruction — Bowel sound ↑
  - GOO
  - SBIO
  - SMA syndrome
- Motility disorder — Bowel sound ↓/ N
  - paralytic ileus
  - gastroparesis
  - colonic pseudo-obstruction
  - achalasia
- Hepato-biliary-pancreatic causes — Bowel sound ↓/ N
  - acute hepatitis
  - acute cholecystitis
  - acute pancreatitis
- Bowel inflammation — Bowel sound ↓
  - acute mesenteric ischemia
  - gastroenteritis
- Pharyngeal irritation — Bowel sound N
  - Eating disorder

Non-GIT — Bowel sound ↓/ N

- Peritoneal irritation
  - peritonitis, carcinomatosis
- CNS/ PNS
  - Increase intracranial pressure
  - CNS infection
  - Acute stroke
  - Others: vestibular causes, migraine
- Cardiac
  - AMI, heart failure
- Systemic
  - Metabolic (e.g. DKA, Addisonian crisis, hypoNa, uraemia)
  - Drugs (e.g. chemotherapy, metformin, colchicine, Levodopa, anticonvulsants, antibiotics, narcotics, digoxin)
  - Hyperestrogenemia (pregnancy, OCPs)
  - Alcohol abuse


Dilated bowel
Valvulae conniventes
R
R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BAS
A
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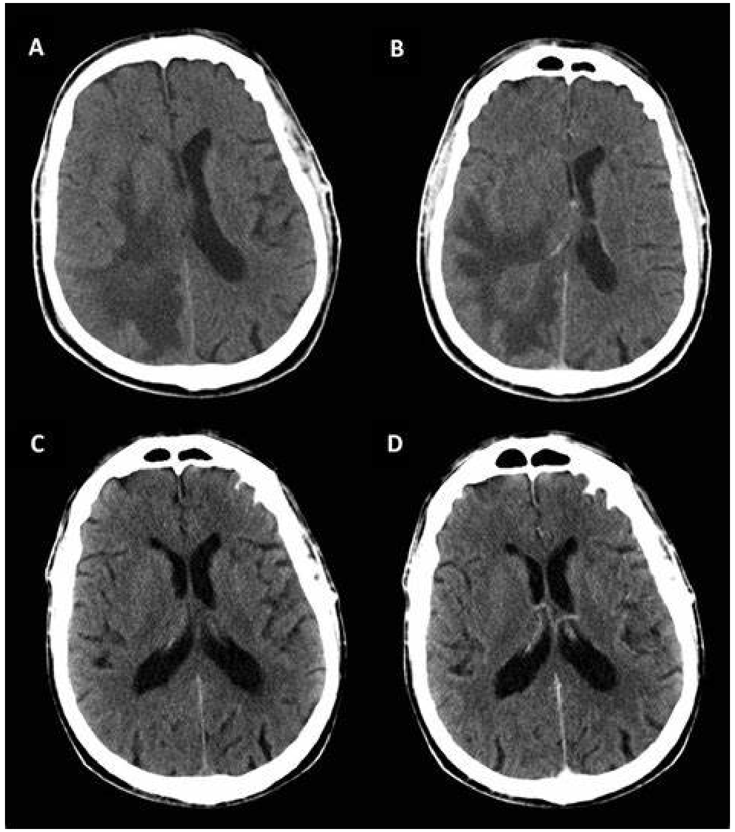
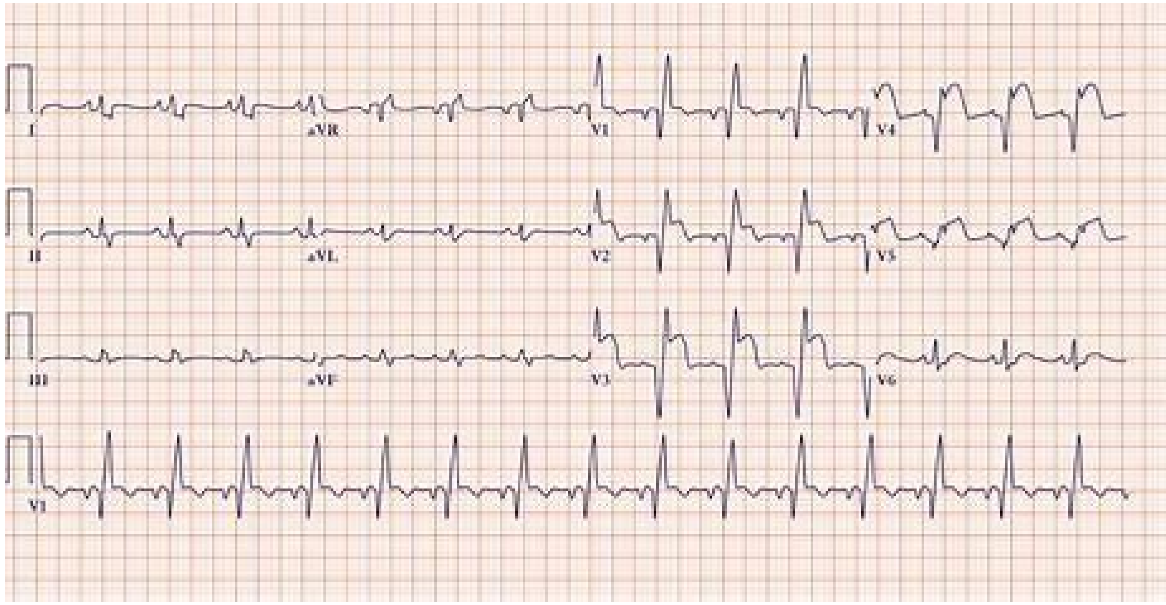


Sample Type:_______ART
ON AIR
At 37 C
pH_________________ 7.20
PO2________________13.1 kPa
PCO2_______________2.5 kPa
BE_________________-9.0 mEq/L
HCO3_______________15.0 mEq/L
Lactate____________3.1 mmol/L
Glucose ___________25.1 mmol/L

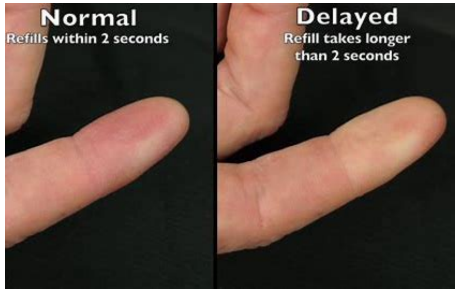


Salient clinical features at history taking

- Onset (acute – arbitrarily defined as duration <1 week vs chronic)
- Timing of vomiting (empty stomach vs. post-prandial)
- Nature of vomitus (total undigested vs. acidic vs. bilious)
- Preceding nausea (absent: i.e. projectile vomiting, more suggestive of direct stimulation of emetic center)
- Constitutional symptoms
- Associated GI tract symptoms (e.g. abdominal pain, jaundice, diarrhea, hemetemesis)
- Associated CNS/ vestibular/ cardiac symptoms
- LMP for female

- Past medical history (e.g. DM: medication? DKA? Gastroparesis? Stroke? MI?; Psychiatric history)
- Drug history
- Surgical history (e.g. abdominal surgery and adhesive SBIO)

Physical examination for vomiting



- General exam
  - Vitals: is patient in hypovolaemic shock?
  - Hydration status: capillary refill, mucous membrane, skin turgor, body weight, urine
  - Acidotic breathing (metabolic acidosis)
  - Jaundice? Uraemic complexion?
  - Russell's sign (callused knuckles of hand)

- Abdominal exam
  - Inspection: distension, surgical scars, hernia
  - Palpation: peritoneal signs, Murphy's sign, abdominal mass
  - Percussion: characterize abdominal mass, look for shifting dullness
  - Auscultation: bowel sound!!!
- Targeted exam for other systems based on what you obtained from your history
  - E.g. Cardiovascular exam
  - E.g. Neurological + fundi exam

Workup for acute vomiting

- Urinary pregnancy test for women of child-bearing age
- CBC: leukocytosis, reactive thrombocytosis, anaemia
- LFT, GGT: hepato-biliary causes, hypoalbuminaeia
- Amylase (+/- lipase): acute pancreatitis, IO, bowel ischemia, perforated gut
- RFT, CaPO4: Cr $\uparrow$, Ca$\uparrow$, electrolyte disturbances (particularly K$\downarrow$, Na $\downarrow$ or Na$\uparrow$), AKI
- Random glucose: DKA
- Ketone: DKA
- Venous blood gas: metabolic acidosis, metabolic alkalosis
- Serum ethanol level
- ECG: U wave, flat T wave, long QTc, myocardial ischemia
- Cardiac enzymes
- CXR: aspiration pneumonia, perforated viscus, Boerhaave's syndrome
- AXR: dilated small or large bowel, gastric distension
- Imaging: USG or CT abdomen; CT brain

words underlined refer to complications from vomiting; those not underlined refer to causes of vomiting

Workup for chronic vomiting

- ***A detailed clinical history and physical examination*** are central to narrowing down the diagnoses
- ***Upper GI endoscopy***
- Contrast studies
- ***Gastric emptying study***
- Cross-sectional imaging
- CT or MR enterography
- Referrals (e.g. neurology, psychiatry)

Management of acute vomiting

1. Fluid rehydration (chart I/O)
2. Correction of electrolyte disturbance
3. Nutritional support (enteral vs. parenteral)
4. Pharmacological therapy: commonly used ones are listed below
   - Central anti-emetic agents
     - Dopamine D2 receptor antagonist:
       - Metoclopramide iv or oral (Maxolon 10mg q8h): S/E: somnolence, extrapyradmial effects
       - Domperidone oral only (10mg TDS): cross BBB poorly, so less extrapyramidal S/E, risk of increase QTc
         Both can increase prolactin
     - Antihistamines and antimuscarinic agents (e.g. diphenhydramine): S/E drowsiness, C/I in glaucoma, asthma
     - Serotonin antagonists (e.g. ondansetron)
     - Neurokinin-1 receptor antagonist (e.g. Aprepitant)
   - Peripheral prokinetic agents
     - Serotonin 5HT4 receptor agonist: metoclopramide
     - Motilin receptor agonists: erythromycin iv (3mg/kg q8h): S/E long QTc ← more specific for gastroparesis

5. Look for and treat the underlying cause

Overview of gastric motility problems: Gastroparesis

Definition & Clinical presentation of gastroparesis

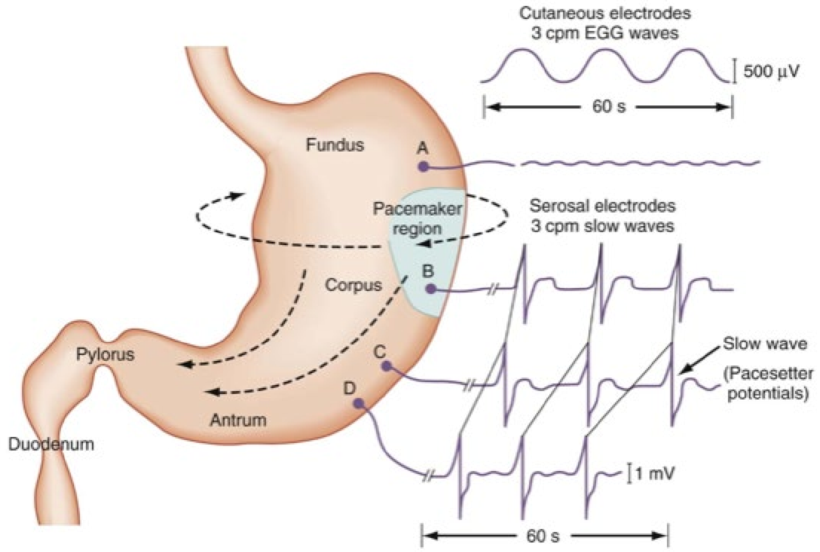
- A syndrome of:
  - ***Delayed gastric emptying***
  - In the ***absence of mechanical obstruction***
  - Cardinal symptoms of early satiety, post-prandial fullness, nausea, vomiting, PPI-refractory GERD (less specific: bloating, upper abdominal pain)
- Prevalence:
  - Among type 1 DM: 5-40%
  - Among type 2 DM: 1-20%

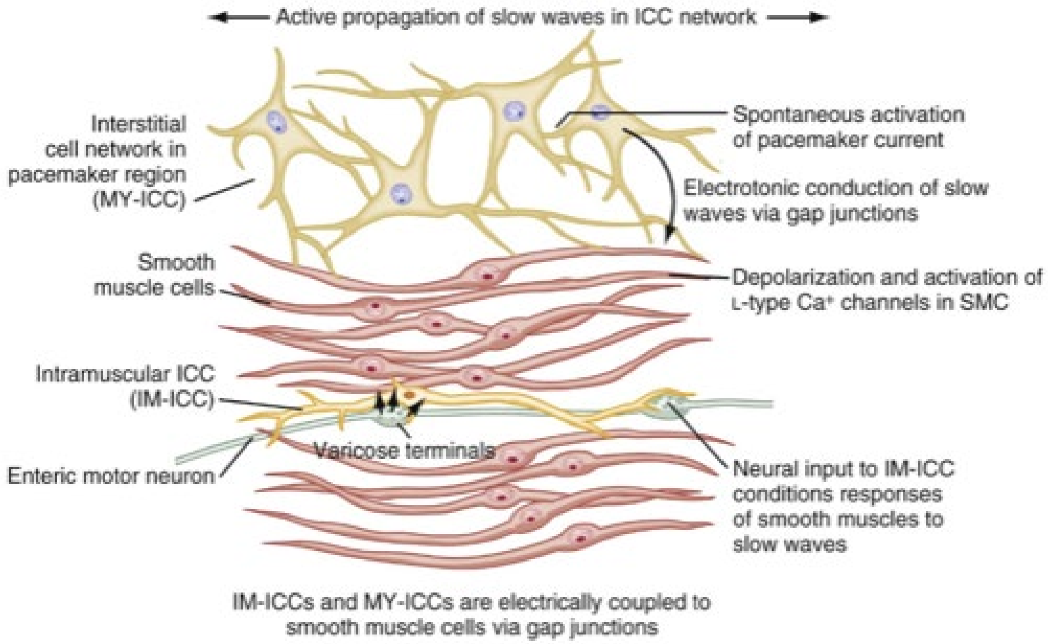
  ‘**Diabetic gastroparesis**’
  - Non-diabetic controls: 0.2%

  ‘**Idiopathic gastroparesis**’

Pathophysiology of gastroparesis

- Stomach is a sophisticated sphere of smooth muscle organized into circular, longitudinal and oblique muscle layers
- ***Gastric myoelectrical activity*** originates from the pacemaker region on the greater curvature of stomach (between fundus and proximal corpus): Interstitial Cells of Cajal (ICC) = pacemaker cells ← from c-Kit positive mesenchymal cell precursors
- Slow waves occur at ~3 cycles per minute





- Depletion of ICCs in the corpus-antrum

Causes of gastroparesis

- Diabetes mellitus
- Idiopathic
- Thyroid dysfunction
- Neurological disease (e.g. Parkinsonism, amyloidosis, paraneoplastic disease)
- Post-surgical (e.g. vagotomy, fundoplication, bariatric surgery)
- Autoimmune disorders (e.g. scleroderma)
- Medication (e.g. GLP-1 agonist, narcotics, anti-cholinergics, cyclosporine)

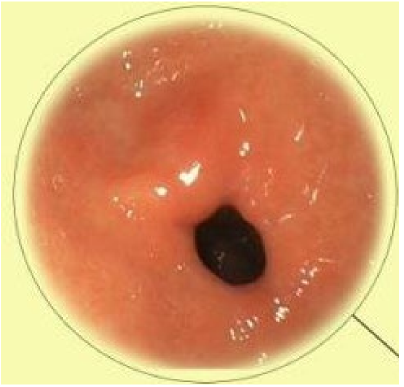
Triggering factors for gastroparesis

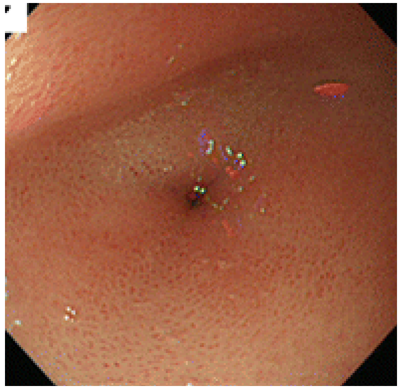
- Infection (e.g. post-viral)
- Hyperglycaemia
- Ischemia
- Electrolyte disturbance

DDX for gastroparesis

- Cyclic vomiting syndrome
- Rumination syndrome
- Eating disorders
- Cannabinoid misuse

Diagnostic approach + workup



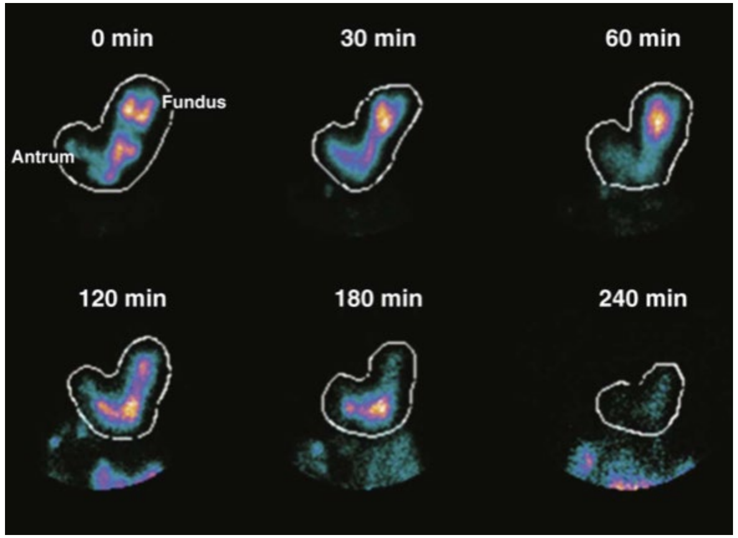


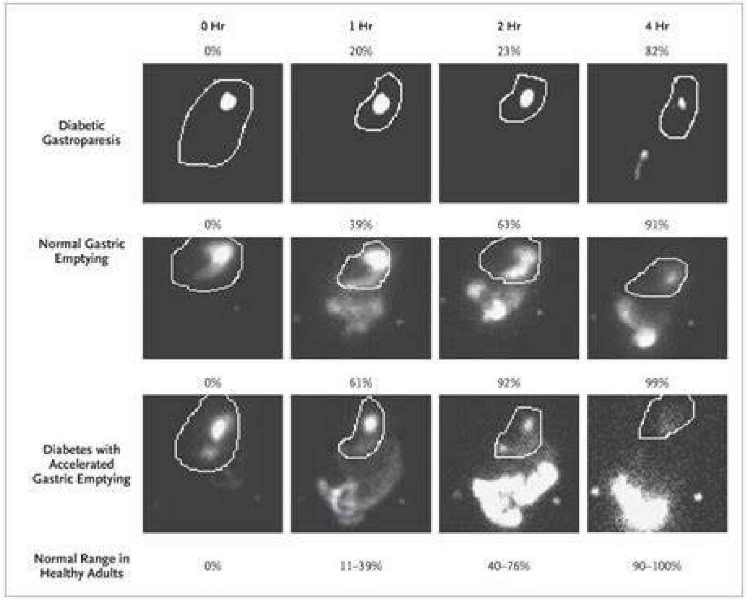
1. ***<u>Exclude gastric outlet obstruction</u>***
   → endoscopy +/- cross-sectional imaging
2. Scintigraphy
   → ***<u>gastric emptying study</u>*** (solid meal)
3. Look for secondary causes and ***<u>precipitating factors</u>***, and correct the reversible ones
   →Review history: viral prodrome, drug list, past medical/ surgical history
   →CBC, RG, LRFT, CaPO4, TFT, blood gas
4. Prepare for ***<u>pharmacological treatment</u>***
   → ECG
5. ***<u>Nutritional support</u>***
   → enteral route (post-pyloric feeding preferred) preferred over parenteral route
   → NG tube for decompression; NJ tube for temporary feeding
   → PEG-J: percutaneous endoscopic gastrostomy with jejunal extension tube
   → Jejunostomy +/- gastrostomy for venting

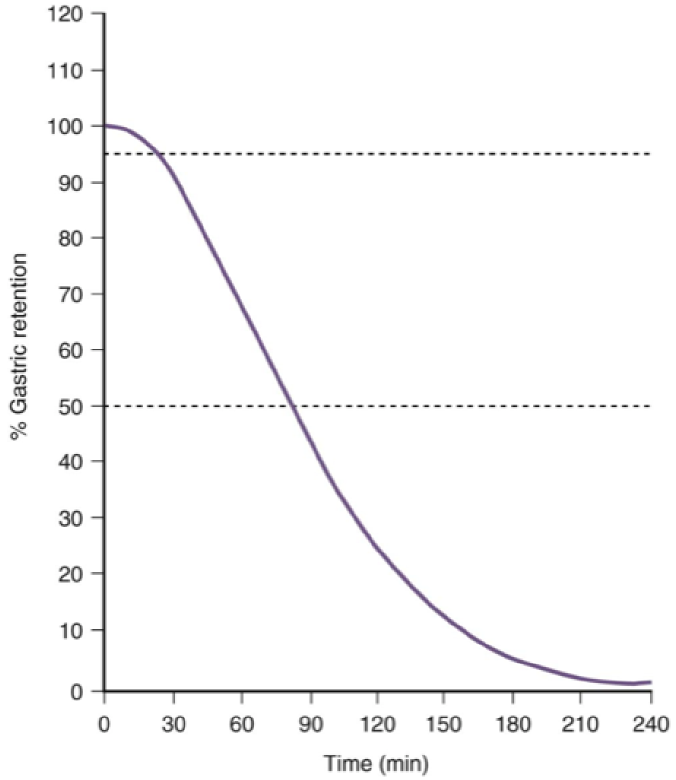
Gastric emptying study (scintigraphy)

<u>Precautions</u>

- Withhold drugs that affect gastric motility for 48-72 hours: narcotic opioid analgesics, anticholinergic, metoclopramide, macrolides, domperidone
- Control glucose to below 15 mmol/L
- Solid phase meal with a 99mTc sulfur colloid-labeled egg sandwich as a test meal or EggBeaters (jam + toast + water)
- Standard imaging at 0, 1, 2 and 4 hours
- ***<u>Normal <10% retention at 4 hours</u>***







Therapy for gastroparesis

- ***<u>Prokinetic agents</u>***
  - Metoclopramide (10mg TDS PO or Q8H IV) – dopamine receptor antagonist
    - <1% risk tardive dyskinesia
    - prolong QTc
  - Domperidone (10mg TDS PO) – dopamine receptor antagonist
    - less CNS S/E than metoclopramide
    - hyperprolactinaemia
    - prolong QTc; avoid if QTc >470ms for male or >450ms for female
  - Erythromycin (IV 3mg/kg q8H) – motilin receptor agonist
    - prolong QTc
    - tachyphylaxis due to down regulation of motilin receptor

- ***<u>Gastric electrical stimulation</u>***
  - marginal benefit – mainly in patients with diabetic gastroparesis

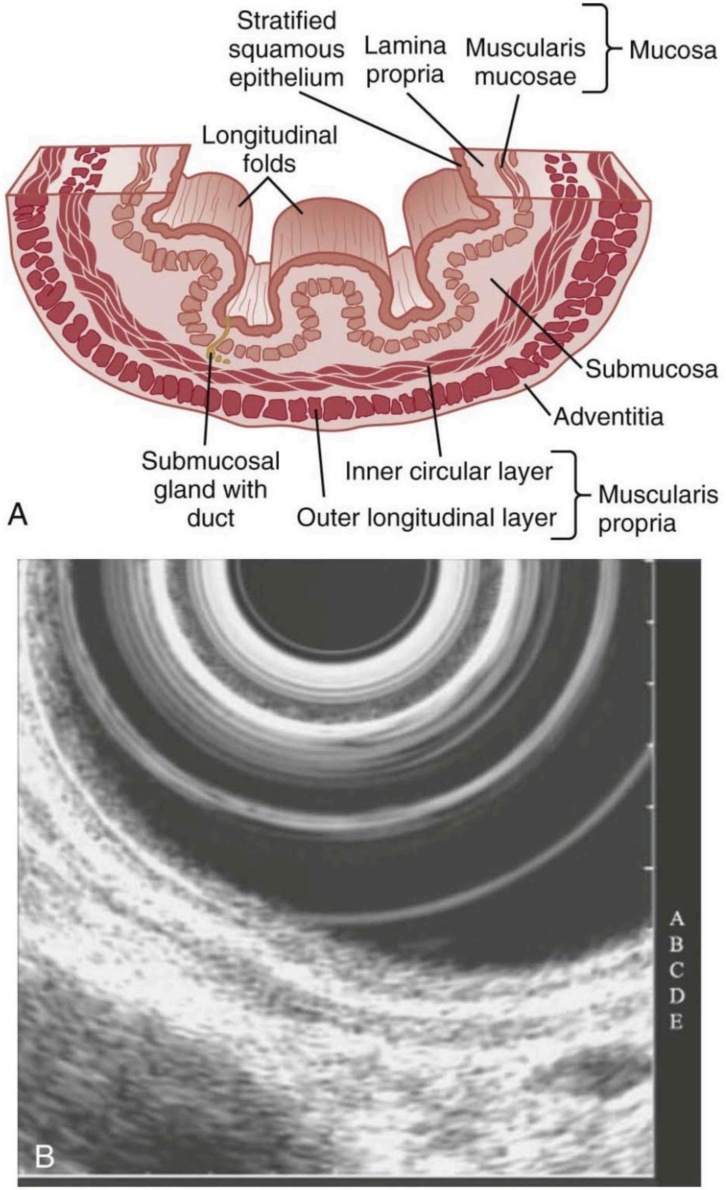
- Gastric peroral endoscopic myotomy (G-POEM)
  - modest benefit

No use: intra-pyloric botulinum toxin injection

Overview of benign esophageal lesions

General considerations

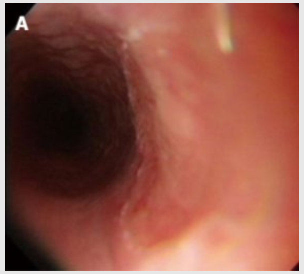
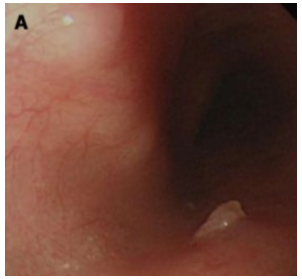
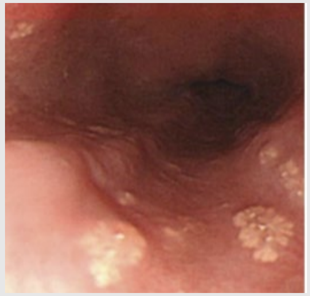
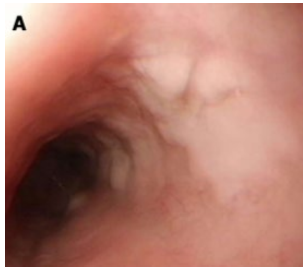
- ***<u>Mostly asymptomatic</u>***, incidentally found during endoscopy for other reasons
- Rare, prevalence ≤0.5%
- ***<u>Epithelial lesions vs. sub-epithelial lesions</u>***
- ***<u>Endoscopic ultrasound</u>*** is helpful for sub-epithelial lesions
- Biopsies maybe helpful sometimes



A, C, E are hyperechoic
B, D are hypoechoic

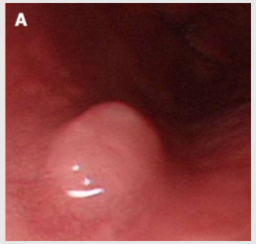
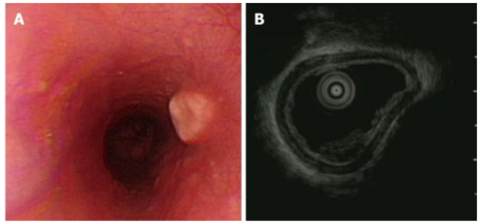
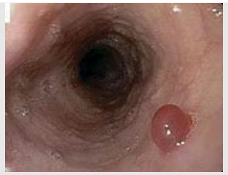
A: mucosa/interface with lumen, B: muscularis mucosae, C: submucosa, D: muscularis propria, E: adventitia

Epithelial lesions

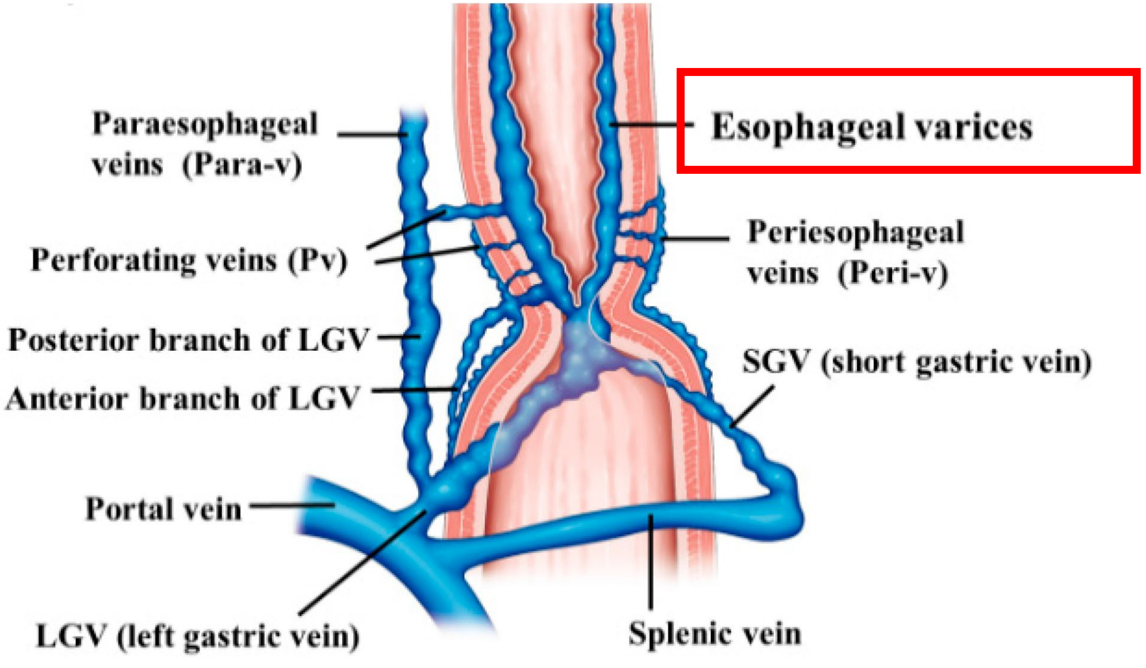
Condition	Incidence	Appearance	Remarks
Heterotopic gastric mucosa	0.1-10%		Differentiate from Barrett's esophagus)
Squamous cell papilloma	0.01-0.45%		Hypothesis: chronic mucosal irritation and infection with HPV
Xanthoma	Unknown Ectopic xanthoma in GIT: stomach (75%), esophagus (12%), duodenum (12%)		Not necessarily associated with hypercholesterolaemia Hypothesis: derived from focal mucosal damage, and lipids from broken down cell membranes captured by interstitial histiocytes
Acanthosis glycogens	Unknown		nil

Tsai SJ et al. World J Gastro 2015

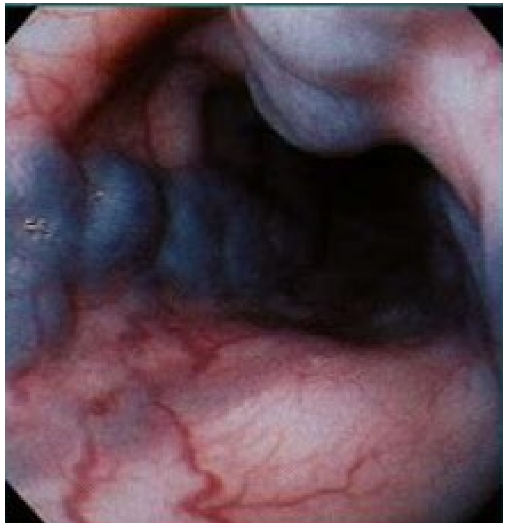
Sub-epithelial lesions

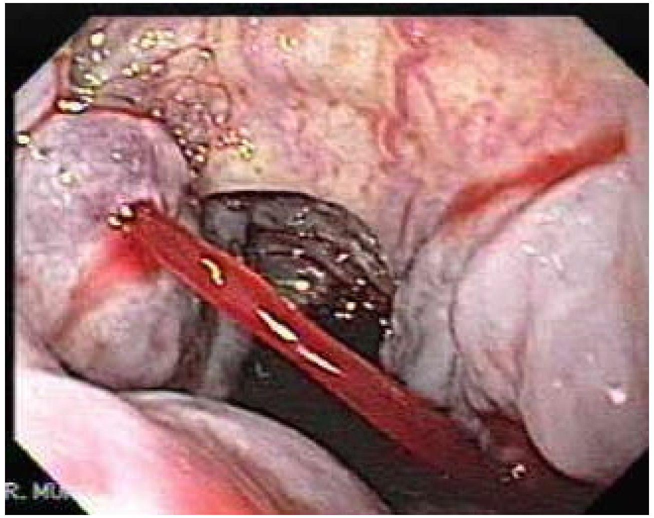
Condition	Incidence	Appearance	Remarks
<u>Leiomyoma</u>	Most common benign esophageal neoplasm (2/3 of all tumors in this region)		Arise from the muscularis propria layer of esophageal wall Can cause dysphagia Malignant transformation rare
Granular cell tumor	2nd most common subepithelial tumor in esophagus		Arise from the submucosa of esophageal wall (Schwann cell) 1-3% can be malignant
Hemangioma	0.04%		Differentiate from esophageal varices
Carcinoid tumor	Rare; More often in small bowel (accounts for 25% tumor in SB) and stomach	Smooth polypoid-like	Arise from mucosa Chromogranin and synaptophysin +ve Maybe hormone secreting
GIST (gastrointestinal stromal tumor)	Extremely rare Most common in stomach and small intestine		Originate from ICC (i.e. muscle layer) KIT mutation Positive for c-KIT (CD117) All GISTs have malignant potential Difficult to differentiate from leiomyoma

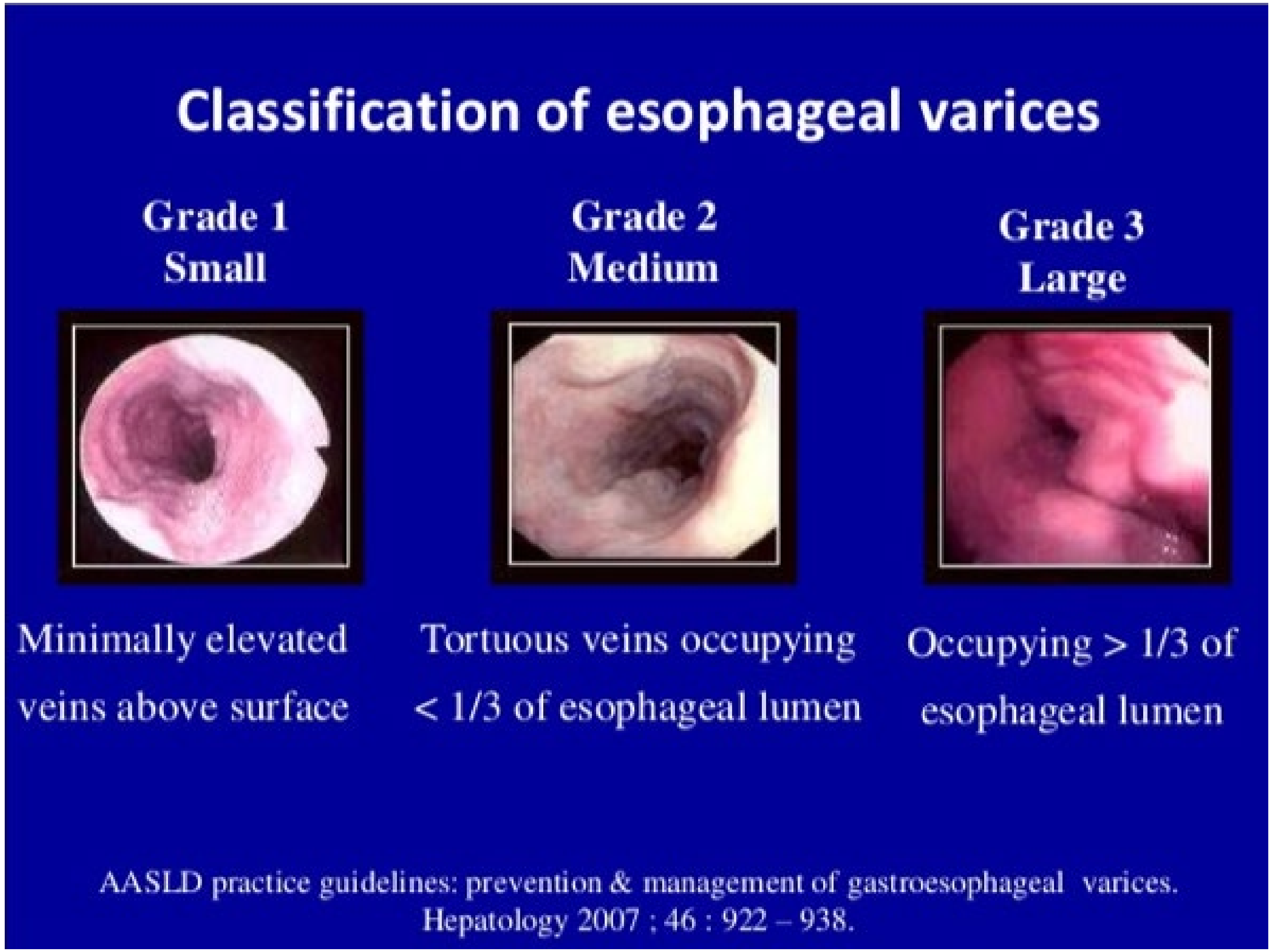
Tsai SJ et al. World J Gastro 2015, Menon L et al. Ther Adv Gastro 2014



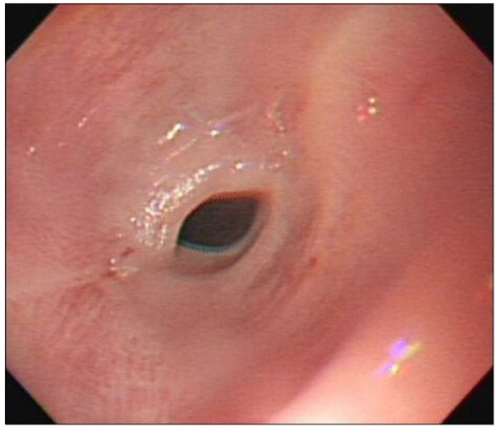
esophageal varices appear as tortuous, dilated blue veins running along the long axis of the esophagus







Others

Condition	Incidence	Appearance	Remarks
<u>Zenker's diverticulum</u> • A pharyngeal pouch diverticulum • Formed by posterior herniation of esophageal mucosa into the Killian's triangle – area of least resistance above the cricopharyngeal muscle	0.01-0.11% Increases with age (esp male >70 years old)		DDX for recurrent vomiting/ aspiration pneumonia/ dysphagia Can be treated by surgical/ endoscopic means (diverticulotomy)
Paterson-Brown Kelly syndrome Also known as Plummer-Vinson syndrome Triad of: • dysphagia • iron deficiency anaemia • esophageal webs	Unknown, rare in developed countries Females 40-70 Rarer in Asian		Can be associated with Zenker's diverticulum Increases risk of esophageal cancer

Le Mouel JP et al, N Engl J Med 2017; Won KY et al. Korean J Helicobacter Up Gastrointest Res 2014

Conclusion

- Good history taking for upper GI symptoms
- Invasive tests are not always necessary for **heartburn** and **indigestion**
- Always consider salient causes if red flag symptoms are present
- Nausea and vomiting can result from GI and non-GI causes
- Gastroparesis can be recurrent and reversible
- Esophageal lesions: epithelial and subepithelial
  - Most subepithelial lesions of esophagus are benign in adults
  - EUS is helpful to characterize subepithelial lesions; biopsy is not always needed

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- Sleisenger and Fordtran's Gastrointestinal and Liver Disease (10th edition, published by Elsevier)

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)